Cross-Out Detection and Classification in Handwritten Documents: A Progressive Architecture Study

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The Problem with Cross-Outs in HTR

Handwritten manuscripts often contain **cross-outs**, words the writer struck writer struck through to delete or revise. These strike-throughs are a routine are a routine part of writing and signal that the marked content should be should be **disregarded**.

Standard Handwriting Text Recognition (HTR) systems have no mechanism for distinguishing crossed-out words from legible text. They process struck-through content as ordinary characters, silently including deleted material in the final transcript.

As a result, machine-generated transcripts faithfully reproduce content the writer explicitly wanted **removed**, corrupting the semantic integrity of the document.

What Goes Wrong

Writer crosses out a word

Intentional deletion deletion of content content

HTR processes it as text

No stroke-level discrimination

Deleted content appears in output

Transcript semantics corrupted

Where Cross-Outs Appear

Cross-outs appear in almost every domain that produces handwritten documents at scale, and the volume of material is well beyond what manual post-processing can keep up with.

Historical Archives

Millions of digitized manuscripts contain editorial strike-throughs that must be excluded from scholarly transcriptions.

Medical Records

Physician notes and prescriptions routinely include crossed-out dosages or diagnoses that must never reach downstream systems.

Exam Scripts & Legal Docs

In exam answers and legal contracts, a cross-out marks a deliberate revision, semantically distinct from the valid text surrounding it.

Cross-out style also encodes intent: a single clean line often signals a correction, while a heavy scratch typically signals redaction. For any real-world HTR pipeline, an automatic detector is therefore a critical pre-processing step.

Three Tasks, One Progressive Study

The project is organised around three interdependent tasks. The third evaluates the trained models on real handwriting that was never seen during training.

1

Binary Detection

Task 1: classify each word image as either clean or crossed-out. This establishes the baseline capability of reliably flagging the presence of any cross-out stroke.

2

7-Class Identification

Task 2: if a word is crossed out, identify the type (Single-Line, Double-Line, Diagonal, Cross, Wave, Zig-zag, or Scratch). Style classification adds semantic depth to detection.

3

Zero-Shot Real Evaluation

Task 3: collect a custom real-handwriting dataset and evaluate the trained models on it. The evaluation is strictly zero-shot, with no retraining or threshold-tuning on the custom data.

Core approach: a progressive architecture study. We build three models, each one adding a single new capability on top of the previous, and then test how well they generalise to real handwriting.

Dataset 1: IAM with Synthetic Cross-Outs

Source & Structure

The **IAM Handwriting Database** provides a large-scale, pre-split corpus of isolated word images from real handwriting. Seven synthetic cross-out styles were overlaid programmatically on each word image to produce a labelled multi-class dataset.

Test Split Statistics

162K
Total Test Images

20.3K
Clean Words

142K
Crossed-Out Words

Seven Synthetic Cross-Out Styles

- Single-Line
- Double-Line
- Diagonal
- Cross
- Wave
- Zig-zag
- Scratch

The styles span the range of real cross-out appearances, from light editorial marks to heavy redaction strokes.

Natural ***1 : 7 class*** imbalance *(clean vs. crossed-out) must be explicitly accounted for in model training and evaluation strategies.*

Dataset 2: Custom Paper Dataset (Task 3)

Collection Protocol

To test generalisation, we hand-wrote a custom dataset entirely independent of the IAM corpus. Eight distinct words were written and crossed out in each of the eight cross-out styles, giving **64 word images** on A4 paper, written with a ballpoint pen under daylight and captured with a smartphone.

Words were extracted automatically using **connected-component analysis**, which removes any manual cropping bias.

Design Principles

8 × 8 Grid

8 words × 8 cross-out styles = 64 images covering the full style space.

Realistic Capture Conditions

Ballpoint pen, A4 paper, ambient daylight, smartphone capture. This matches the conditions of real archival digitisation workflows.

Strictly Zero-Shot

This dataset was **never used** in training, validation, or threshold-tuning, so it provides a clean held-out test of generalisation.

Solution: A Progressive Model Architecture

Rather than training three independent models, we build an **engineering progression**. Each model inherits its predecessor's backbone and adds exactly one new capability, so the contributions accumulate rather than being measured in isolation.

Model 1 : Baseline CNN

A convolutional backbone trained for **binary detection** (clean vs. crossed-out). This establishes the feature-extraction foundation that the next two models inherit and adapt.

Model 2 : Multi-Task Learning

Adds a **shared representation** with two output heads, one for binary detection and one for 7-class style identification. Joint training forces the backbone to learn features useful for both objectives at once.

Model 3 : Self-Attention

Augments M2 with a **self-attention module** that captures global stroke trajectories across the full word image. This lets the model reason about long-range spatial relationships that local convolutions cannot.

This is an engineering progression, not a controlled ablation. Each model is built on the last by adapting the previous backbone rather than reinitialising it, so every step is a genuine incremental change.

Model 1: EfficientNet-B0 Baseline

Architecture

ImageNet-pretrained CNN backbone

Two separate models, one per task (binary and 7-class)

New classifier head bolted on top

Class-balanced sampling to handle the 1:7 imbalance

~4M

Trainable Parameters

Model 2: + Multi-Task Learning (MTL)

Backbone warm-started from Model 1's Task 2 checkpoint.

Parallel Head 1: Detection

Binary output (clean vs. crossed-out).

Parallel Head 2: Classification

7-class identification of cross-out types
(Single-Line, Double-Line, Diagonal, Cross, Wave, Zig-zag, or Scratch).

Heads initialised randomly; backbone fine-tuned end-to-end.

Model 3: + Self-Attention

Warm-started from Model 2 (engineering progression).

1. **7×7 Spatial Feature Map**
2. **49 Tokens**
3. **2 Transformer Layers, 8 Heads, Learnable Position Embeddings**

Adds a Transformer attention block on top of Model 2's features

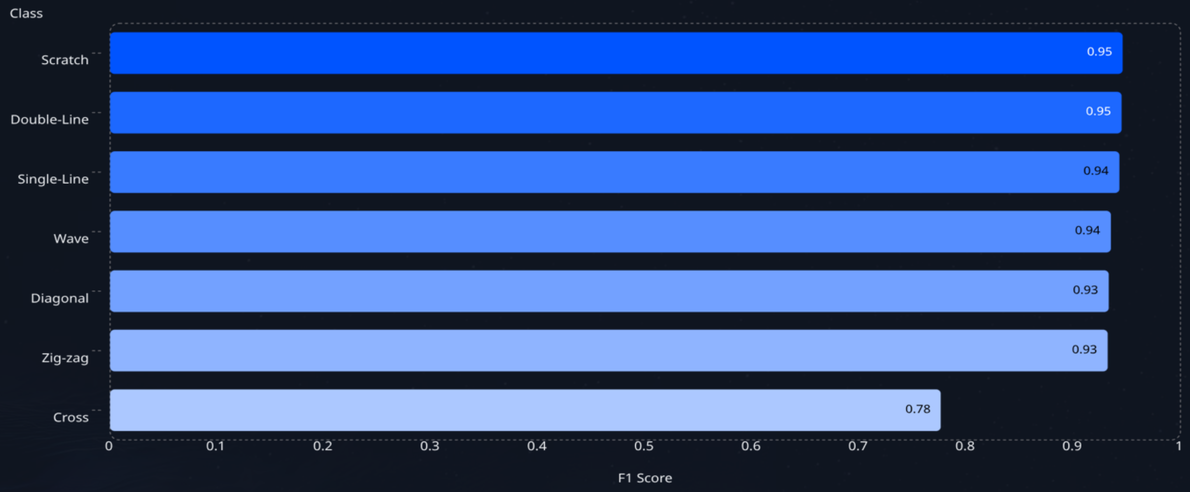
Lets the model see the whole word at once instead of just local patches

A diagonal cross-out spans the entire word, so the model needs the global context that local convolutions cannot provide.

Results: IAM Test Set - Model 3 (CNN + Transformer + MTL)

7-class cross-out type classification on 142,142 held-out IAM samples

Per-class F1 score



Overall accuracy: 90.91%

Six of seven classes above 0.93 F1

"Cross" is the only weak class, its strokes visually overlap many other styles, so the model uses it as a catch-all category.

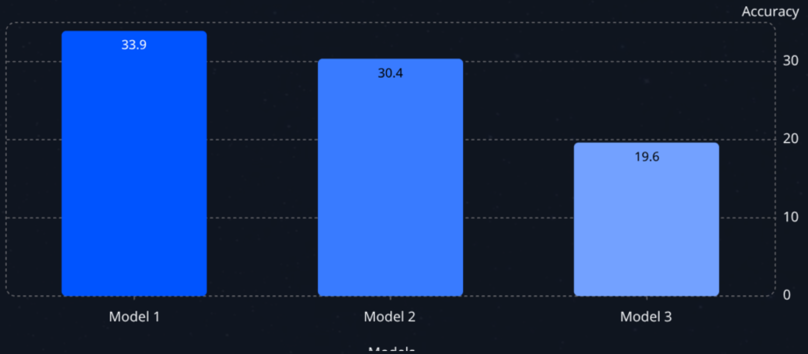
Model 3 trained on IAM with 7 synthetic cross-out styles. Macro-F1 = 0.9156, Macro-Precision = 0.9372.

Results: Custom Paper Dataset - Zero-Shot

All three models, no paper data used in training (strict generalisation test)

Binary AUC-ROC

7-Class Accuracy (%)



Model 3's binary AUC of 0.996 almost matches its IAM IAM performance, showing that attention preserves the preserves the cross-out signal under domain shift.

On zero-shot 7-class accuracy, Model 1 narrowly leads at 33.9%. Its features generalise best when no adaptation is available.

Different models lead on different metrics. Zero-shot accuracy on its own is a poor predictor of overall capability.

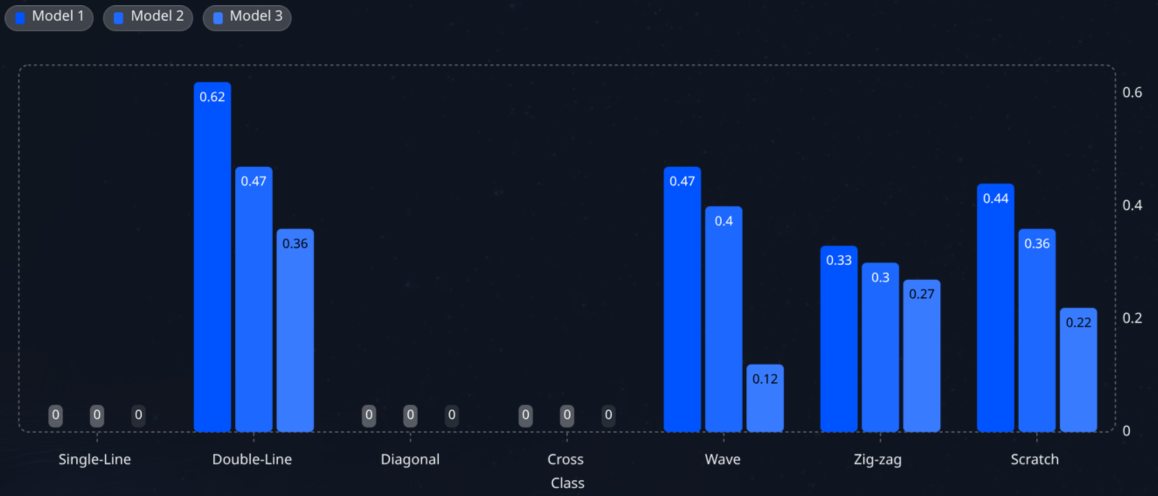
The binary-accuracy paradox: Model 1 reports 87.5% accuracy but its AUC is only 0.484. It is predicting 'crossed-out' for almost every sample and gets credit because 87.5% of the paper samples really are crossed-out. Model 3's 14% accuracy is a threshold-calibration problem rather than a discrimination problem; its ranking quality is near-perfect.

64-image paper dataset · ballpoint pen on A4 · smartphone capture · constructed after training, never used for any tuning

Three Cross-Out Styles Fail in Every Model

Per-class F1 on the paper dataset. Single-Line, Diagonal and Cross are essentially never detected.

Per-class F1 (zero-shot, paper dataset)



Single-Line, Diagonal, and Cross all score F1 = 0 in every model.

These three styles share a structural property: thin, sparse strokes that overlap the underlying letters.

Double-Line, Wave, and Scratch (heavy or repetitive strokes) survive the domain shift because they dominate the image globally.

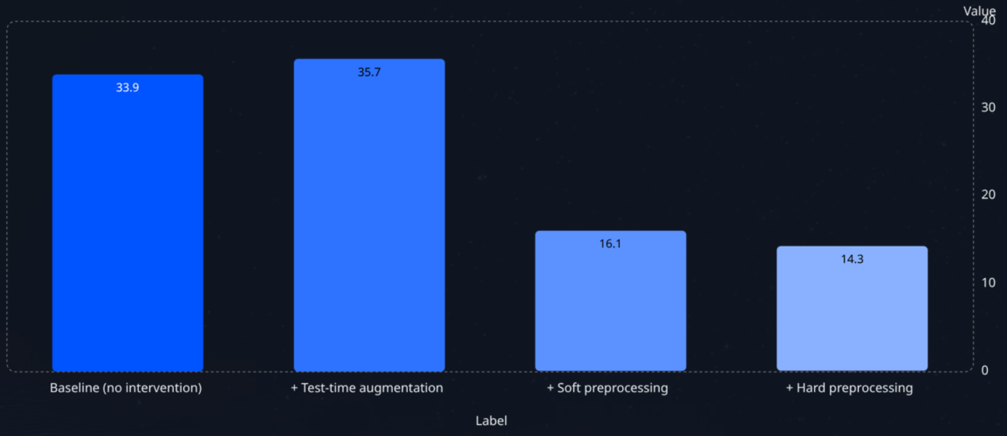
The failure pattern is identical across three independently-trained models, which rules out idiosyncratic training error and points to a feature-level mismatch between synthetic and real strokes.

56 crossed-out paper samples · 8 per cross-out class · strictly zero-shot

Inference-Time Interventions Don't Help

We tried fixing zero-shot performance with preprocessing and test-time augmentation. Only TTA gives a small lift; preprocessing actively hurts.

Model 1 : 7 - class accuracy on paper, by intervention



Best honest zero-shot result: 35.71% with TTA alone (+1.8 pp over baseline).

Scan-like preprocessing (CLAHE + Otsu binarisation) cuts accuracy by about 20 percentage points.

Preprocessing removes the subtle stroke variations the classifier had learned to rely on.

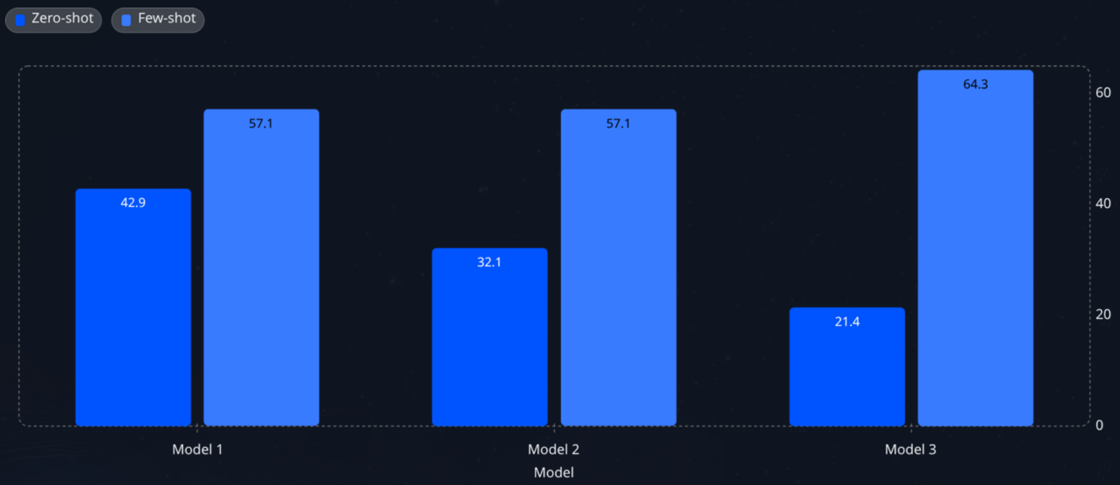
If image quality (noise, contrast, lighting) were the bottleneck, preprocessing would help. Because it actively hurts, the real problem must be a feature mismatch between synthetic and real strokes rather than pixel-level quality.

TTA: 5-view averaging (horizontal flip + small rotations) · Soft preprocessing: CLAHE + denoise + sharpen · Hard preprocessing: soft + Otsu binarisation · Numbers shown for Model 1; Models 2 and 3 follow the same pattern

Few-Shot Adaptation Recovers Most of the Gap

Frozen backbone + only the 7-class head fine-tuned on 4 examples per class

Paper 7-class accuracy: zero-shot vs. few-shot



Just 4 real examples per class transforms zero-shot accuracy from 21–43% to 57–64%

Model 3 gains the most (+42.9 pp) and ends the highest (64.29%)

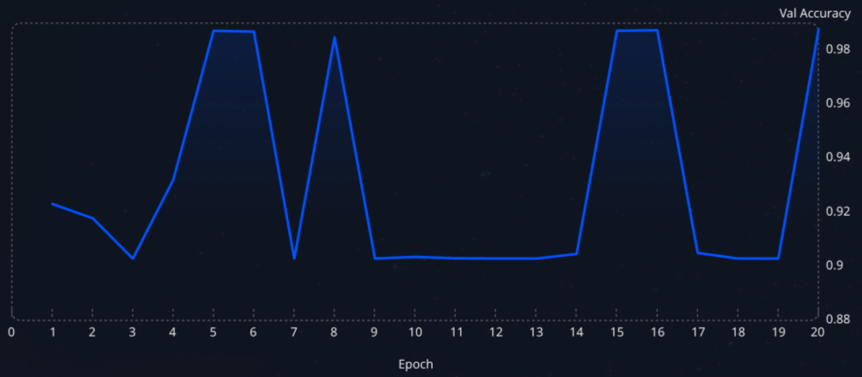
Macro-F1 ordering also flips: M3 was worst zero-shot (0.158), best after adaptation (0.647)

The model that performed worst zero-shot now wins. Zero-shot accuracy alone is a poor predictor of how a model will perform once even a small amount of target-domain data is available.

56 paper samples · stratified split: 4 train / 4 test per cross-out class · seed = 42 · backbone frozen · 30 epochs head training · AdamW lr = 1e-3

What We Learned

Bimodal validation behaviour



0.903 = class-prior collapse (always predict 'crossed-out') · 0.987 = genuine binary classifier

1. **Feature-distribution gap, not capacity gap**
   The same backbone that scored F1 = 0 zero-shot reaches F1 = 0.86 after just 4 real examples per class.

2. **Zero-shot accuracy is a poor deployment proxy**
   Model 3 was the worst zero-shot but became the best after after minimal adaptation. The ranking flips completely.

3. **Attention preserves discriminability under domain shift**
   Model 3 paper AUC = 0.996, almost identical to its IAM performance, global context survives the gap.

4. **Imbalanced binary tasks need pos_weight**
   Without it, the model oscillates between a genuine classifier and a constant-positive predictor as shown in the val curve.

One-line fix for the validation instability: use BCEWithLogitsLoss with pos_weight=7.0. This tilts the loss surface against the class-prior minimum and produces a monotonic curve.

The 0.903 attractor = IAM val set's class prior (always predict 'crossed-out') · 0.987 = genuine binary classifier · Best-checkpoint selection always picks the genuine-classifier mode, so deployed weights are unaffected

Limitations & Future Work

What our results don't tell us, and where we'd go next.

Limitations of this study

Small paper dataset

64 images, one writer, one pen, one lighting condition. We can't claim these results generalise to other writers, inks, or capture setups.

Coarse per-class granularity

8 images per class on the full set, 4 per class for few-shot. Per-class accuracy moves in 12.5% steps, so the absolute numbers are indicative, not statistically tight.

Synthetic-generator parameters undocumented

Our stroke-width-mismatch claim is inferred from failure modes. We did not compare it directly against the generator.

Shared EfficientNet-B0 backbone

A stronger backbone could shift the synthetic-to-real gap, though we'd expect the same qualitative pattern.

Future work

Camera-aware augmentation on IAM

JPEG compression, motion blur, perspective distortion, shadow gradients. Closes the synthetic-to-real gap without any paper training data.

Larger, more diverse paper dataset

Multiple writers, different pen types, varied capture devices and lighting conditions.

Stronger backbone

EfficientNet-B3 or a Swin Transformer, benchmarked against the current synthetic-to-real gap.

Imbalance-aware loss for binary head

BCEWithLogitsLoss(pos_weight=7.0) to fix the class-prior collapse we saw in training (see Slide 16).

We deliberately excluded few-shot fine-tuning, paper-data threshold calibration, and any other paper-data-dependent training from our zero-shot evaluation, so we measure pure generalisation. Adaptation (Slide 15) is shown as a separate complementary experiment.

Conclusion

A progressive architecture study for cross-out detection, and the model that ultimately wins

What we built

Model 1 : EfficientNet-B0 Baseline

Strongest binary classifier on IAM (97.98%)

Model 2 : + Multi-Task Heads

Shared representation between detection and classification. Warm-started from Model 1.

Model 3 : + Self-Attention - "Overall Winner"

Best paper binary AUC (0.996) · Highest few-shot gain (+42.9 pp) · Best IAM macro-F1 (0.9156). Warm-started from Model 2.

Key headline numbers

96.49%

Model 3 binary accuracy on IAM

0.996

Model 3 binary AUC on real handwriting

+42.9 pp

Model 3 gain from 4-example adaptation

Strict

Zero-shot, no paper data used in training